

Voice NFT Studio - Demo Script

Installation

To set up this project, install all required dependencies:

```
npm install --save @solana/web3.js
```

Full dependencies include Solana wallet adapters, ElevenLabs integration, and UI components. After cloning, run:

```
npm install
```

Then start the dev server:

```
npm run dev
```

Demo Script for Judges

1) Opening Remarks

"Hi everyone, I'm [Your Name]. I built a web3-native creative studio that turns spoken ideas into real Solana NFTs in one fluid, composable flow. This is not just a demo — it's a proof that voice, AI, and on-chain provenance can coexist in a modern product with a developer-first architecture.

I'm incredibly passionate about web3 because it isn't just a technology stack, it's a new paradigm for ownership, trust, and creative expression. This project is my answer to that: a live, decentralized experience where the user speaks one sentence and the blockchain records the result."

2) What It Is

"This app is a multimodal pipeline:

- First, I capture voice in the browser.
- Then I transcribe it using ElevenLabs.
- I turn that transcript into a generative image.
- Finally I mint it as a Solana NFT on Devnet with on-chain metadata.

It's engineered to be both intuitive for users and architecturally rigorous for judges who care about composability and real web3 fundamentals."

3) Architecture & Tech Stack

"This is built with:

- React + Vite for a fast frontend developer experience,
- Tailwind CSS for polished, responsive UI,
- Solana wallet adapter + Metaplex UMI for decentralized minting,
- ElevenLabs for speech-to-text and rich voice feedback,
- Pollinations for prompt-driven image synthesis,
- browser localStorage for a lightweight gallery experience.

I intentionally used a data URI metadata pattern for the demo so the minting path stays end-to-end without requiring a server-side file backend. That's a pragmatic, elegant tradeoff for a Devnet prototype."

4) Sponsor Integration Highlights

"I also want to call out the sponsor toolchain integration:

- **ElevenLabs:** The speech intelligence engine here — it powers the entire voice transcription and TTS feedback loop. Users speak naturally, and ElevenLabs converts that to text with high fidelity.
- **Solana:** The crypto substrate: wallet connection, transaction signing, Devnet NFT minting, and immutable provenance. Every artwork minted is a real token on the Solana blockchain with verifiable on-chain metadata.
- **v0:** The design philosophy I applied: composability, modular developer ergonomics, and modern web3 primitives. The component architecture is semantic, reusable, and production-ready.
- **TryNoah.ai:** Influenced the product direction — this is built around conversational generative UX and hyper-personalized AI interaction, which is exactly the kind of experiential layer they champion. The voice-first paradigm is key here."

5) Live Demo Flow

"Let me show you the flow in action:

1. I'll connect my Solana wallet (Phantom / Backpack style).
2. I record a voice prompt like 'a bioluminescent synthwave forest.'
3. ElevenLabs transcribes that in real time.
4. The app generates a visual artwork using Pollinations.ai.
5. Then I mint it as an NFT on Solana Devnet.
6. Finally we can inspect the transaction signature and gallery entry.
7. All mints are stored in a beautiful, indexed gallery page."

6) Smart Technical Details to Mention During Demo

- "This is on Solana Devnet, so the user is minting with a real blockchain signature and a real public key."
- "The NFT metadata is encoded as a base64 data URI, which avoids extra server infrastructure for this demo and keeps the asset pipeline pure."
- "I used a wallet-agnostic adapter approach so this works with Phantom, Backpack, and other Solana wallets — no vendor lock-in."

- "The app is intentionally latency-tolerant: the user sees progress states for recording, transcription, image generation, and then minting."
- "Error messages linger longer (8+ seconds) so users can read detailed feedback — I've added console logging to help diagnose any issues with the Pollinations API."
- "Each minted NFT includes the original voice prompt, timestamp, and a direct link to the verified transaction on Solana Explorer."
- "The gallery is on a separate page for a clean, focused UX — users can browse their entire portfolio without clutter."

7) Closing Remarks

"In short, this demo is about bringing AI and web3 together in a seamless human experience. I wanted the judges to see:

- **Real blockchain ownership:** Every mint is a real token with immutable on-chain proof.
- **Real AI intelligence:** ElevenLabs transcription and Pollinations generation work instantly.
- **A product that feels like the next layer of the internet:** Voice-to-NFT in seconds, no friction.

I believe this architecture — combining modular AI APIs with decentralized ownership primitives — is the future of creative tools. Thank you — I'm happy to dive deeper into the architecture, the sponsor integrations, or the on-chain minting logic next."

Key Features Recap

- ✓ Voice recording and ElevenLabs transcription
- ✓ AI prompt-to-image generation via Pollinations
- ✓ Solana wallet integration (Phantom, Backpack, etc.)
- ✓ Real NFT minting on Solana Devnet with Metaplex UMI
- ✓ On-chain metadata stored as data URI
- ✓ Local gallery with transaction history
- ✓ Mute/unmute TTS feedback
- ✓ Error messages that linger long enough to read
- ✓ Console logging for debugging
- ✓ Responsive design with Tailwind CSS

Environment Setup

Create a `.env.local` file in the project root:

```
VITE_ELEVENLABS_API_KEY=your_elevenlabs_api_key_here
```

If `VITE_ELEVENLABS_API_KEY` is not provided, the app will use browser TTS fallback for voice feedback.

Build & Deploy

```
npm run build  
npm run preview
```

The build output is optimized and ready for deployment.

Built with passion for web3. Let's create on-chain. 🌸